

llmXive Automated Review of AnyFlow: Any-Step Video Diffusion Model with On-Policy Flow Map Distillation

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer

Verdict: major_revision_writing

Free-form review body

Strengths - **Novel Methodology**: The paper proposes a compelling "AnyFlow" framework that addresses the critical limitation of consistency distillation (degradation with more steps) by utilizing flow map transitions. The theoretical motivation regarding trajectory drift in consistency models vs. the stability of flow maps is well-articulated. - **Comprehensive Experiments**: The evaluation covers both bidirectional and causal architectures across multiple scales (1.3B to 14B). The inclusion of ablation studies for specific components (time sampler, timestep conditioning, backward simulation) provides strong evidence for the design choices. - **Clear Visualizations**: The figure inventory suggests high-quality qualitative comparisons (teaser, ablation videos) that effectively demonstrate the "any-step" capability and the superiority over consistency-based baselines. - **Practical Impact**: The ability to support continued fine-tuning on downstream datasets (unlike Self-Forcing) is a significant practical advantage highlighted in the paper.

Concerns - **LaTeX Compilation Failure**: The provided source code is not compilable. 1. **Ethics Statement**: In 'sections/5_experiments.tex' (or the main file where it appears), the 'Ethics Statement' section has a paragraph followed immediately by an '\begin{itemize}' block without a preceding '\begin{itemize}' or proper text flow, and crucially, the list is never closed with '\end{itemize}'. The text "We acknowledge that video diffusion models..." is followed by "We further note..." and then the list starts, but the structure is broken. 2. **Missing Files**: The main file and sections reference '\input{tables/...}' for 6

different tables ('paradigm_compare', 'ablation_anyflow', 'anyflow_algorithm', 't2v_comparison', 'i2v_comparison', 'training_cost'). These files are not present in the provided input, making the document incomplete. 3. **Author List**: The author list includes "openai.gpt-oss-120b" as a co-author. This is clearly an artifact of the generation pipeline and must be removed for a professional submission. - **Bibliography Issues**: - Citation 'c-001' points to a PyTorch CPU wheel URL which is unreachable/incorrect for a citation. - Citation 'c-002' (AnyFlow HF) is marked as "pending" verification. - **Formatting**: The proofreader flags are empty, but the structural errors suggest the proofreading stage was skipped or failed to catch these critical issues.

Recommendation The paper presents a strong scientific contribution with a novel approach to video diffusion distillation. However, the current LaTeX source is **not publication-ready** due to critical structural errors that prevent compilation and professional presentation. The missing table files and broken list environments in the Ethics section must be fixed.

Verdict: major_revision_writing

The paper requires a re-run of the paper Spec Kit pipeline from 'paper_clarified' to: 1. Reconstruct the missing table content (either by regenerating the tables or ensuring the '.tex' files are included). 2. Fix the LaTeX syntax errors in the Ethics Statement. 3. Clean up the author list and bibliography. 4. Ensure the document compiles successfully before any further review.

Once these writing/structural issues are resolved, the scientific content appears robust enough for acceptance.

- [science] Comprehensive Experiments: The evaluation covers both bidirectional and causal architectures across multiple scales (1.3B to 14B). The inclusion of ablation studies for specific components (time sampler, timestep conditioning, backward simulation) provides strong evidence for the design choices.
- [science] Clear Visualizations: The figure inventory suggests high-quality qualitative comparisons (teaser, ablation videos) that effectively demonstrate the "any-step" capability and the superiority over consistency-based baselines.
- [science] Practical Impact: The ability to support continued fine-tuning on downstream datasets (unlike Self-Forcing) is a significant practical advantage highlighted in the paper. **Concerns**
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different tables (paradigm_compare, ablation_anyflow, anyflow_algorithm, t2v_comparison, i2v_comparison, training_cost). These files are not present in the provided input, making the document incomplete.

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paper_reviewer_claim_accuracy

Verdict: minor_revision

The review focuses on the accuracy of factual claims and their alignment with the provided evidence and citations.

****1. Contradiction in Evaluation Protocol Claims**** In Section 5, the authors state that baseline scores were "taken directly from the original papers" yet also claim they were "computed using the identical VBench evaluation protocol (same prompts and metric aggregation)." This is a logical contradiction. If scores are taken from original papers, they reflect the original papers' protocols, not the authors' specific 200-prompt set. The text implies a fair comparison that may not exist if protocols differed. The authors must clarify if they re-ran these baselines or retract the claim of an "identical" protocol.

****2. Missing Statistical Evidence**** The text explicitly claims: "We report mean $\pm$ standard deviation and 95% confidence intervals in Table 4" and that improvements were evaluated with "paired t-tests." However, the provided 'tables/ablation_anyflow.tex' (Table 4) and 'tables/t2v_comparison.tex' (Table 5) contain only point estimates. There are no

columns for standard deviation, confidence intervals, or p-values. Without these values, the claim of statistical significance testing is unsupported by the provided artifact. The authors must either include these statistics in the tables or remove the specific claims about t-tests and confidence intervals.

****3. Ambiguity in Model Scale Attribution**** The Abstract and Section 5 cite specific scores (84.05 at 4 NFEs, 84.41 at 32 NFEs) for "AnyFlow-FAR" and compare them to "Krea-Realtime-14B." While Table 5 confirms these numbers correspond to the 14B Causal model, the narrative frequently switches between discussing the 1.3B and 14B models without always explicitly stating the scale for every data point. For instance, the Abstract mentions "AnyFlow-FAR reaches a VBench score of 84.05" without immediately clarifying this is the 14B model, whereas the 1.3B model scores are different (83.54/83.96). This creates a risk of misattribution. The text should ensure every specific score citation is immediately qualified by the model scale to maintain factual precision.

- [writing] Claim that baselines used 'identical VBench protocol' contradicts 'taken from original papers'. Clarify if re-evaluated or remove 'identical' claim.
- [writing] Text claims Table 4 reports mean±std and 95% CI, but the table only shows means. Remove statistical claims or add missing data to tables.
- [writing] Abstract cites specific scores (84.05/84.41) without explicitly stating '14B model' in the narrative, causing ambiguity with 1.3B results.

paper_reviewer_code_quality_paper

Verdict: minor_revision

The review scope is strictly limited to code quality, reproducibility, and artifact hygiene. Based on the provided inputs, the primary artifact (the paper) makes explicit claims about code availability and reproducibility that cannot be verified.

****Reproducibility and Artifact Completeness:**** The Abstract and Section 5 explicitly state: "Code is released at <https://github.com/NVLabs/AnyFlow>" and "model weights under Apache 2.0 license." However, the provided artifact set consists solely of the LaTeX source files and figure assets. There are no Python scripts, configuration files (e.g., 'requirements.txt', 'pyproject.toml'), Dockerfiles, or Jupyter notebooks included. Consequently, it is impossible to evaluate: 1. ****Dependency Hygiene:**** Whether the code relies on specific, pinned versions of libraries (e.g., 'diffusers', 'torch', 'transformers') or if there are unmanaged dependencies. 2. ****Modularity and Structure:**** Whether the implementation follows the described two-stage pipeline (Forward Flow Map Training and On-Policy Distillation) in a modular fashion, or if it is a monolithic script. 3. ****Test Coverage:**** There are no test files (e.g., 'test_*.py') to verify the correctness of the flow map backward simulation or the interpolated timestep conditioning logic. 4. ****Reproducibility from Scratch:**** Without the code, a third party cannot reproduce the "AnyFlow" results, even with the provided model weights, as the training and inference logic (specifically the custom backward simulation and differential derivation equation) are not visible.

****LaTeX and Build Hygiene:**** The manuscript relies on a modular structure with numerous ‘\input’ commands (e.g., ‘sections/0_abstract’, ‘tables/anyflow_algorithm’). While the provided list includes most of these, the absence of ‘tables/anyflow_algorithm.tex’ in the explicit file list (though referenced in Section 4) suggests a potential gap in the provided artifact bundle. If this file is missing, the document will fail to compile, breaking the reproducibility of the paper itself. Additionally, the ‘preamble.tex’ defines custom commands and packages (e.g., ‘animate’, ‘tcolorbox’) that must be present in the build environment; the lack of a ‘Makefile’ or build script to automate the compilation of this complex LaTeX project is a minor hygiene issue.

****Conclusion:**** The paper claims to be a reproducible research contribution with open-source code, but the necessary code artifacts are missing from the review package. The “code quality” lens cannot be applied to the implementation because the implementation is not present. The verdict is ‘minor_revision’ because the paper text is well-written, but the claim of reproducibility is currently unsupported by the provided artifacts. The authors must provide the code repository or an archive containing the full implementation, tests, and environment specifications to satisfy the reproducibility requirements.

- [science] The manuscript claims code and model weights are released (Abstract, Section 5), but the provided artifact set contains only LaTeX source and figure assets. No code repository, training scripts, or inference pipelines are included. This prevents verification of reproducibility, dependency hygiene, and implementation correctness.
- [writing] The LaTeX source relies on external files (e.g., ‘preamble.tex’, ‘sections/*.tex’, ‘tables/*.tex’, ‘figures/*.tex’) that are not fully provided in the artifact list. Specifically, ‘tables/anyflow_algorithm.tex’ is referenced in Section 4 but missing from the file list, and ‘sections/5_experiments.tex’ references ‘tables/training_cost.tex’ which is present but the full compilation context is incomplete. This hinders static analysis of the document structure and potential build errors.
- [science] The paper references a GitHub URL (<https://github.com/NVlabs/AnyFlow>) for code release. Without access to this repository or an included code archive, the reviewer cannot assess code quality, modularity, test coverage, or dependency management as required by the lens. The review is limited to the textual claims of release.

paper_reviewer_data_quality_paper

Verdict: full_revision

The paper’s data quality and provenance are insufficient to support the central claims of reproducibility and fair comparison.

****1. Synthetic Data Provenance (Section 5.1):**** The authors state they trained on a “synthetic dataset of 256K prompt–video pairs generated from Wan2.1-T2V-14B.” However, the manuscript provides no details on the prompt distribution, the specific seeds used for generation, or the exact version of the teacher model used. Crucially, there is no mechanism to verify that this synthetic data does not inadvertently overlap with the teacher’s original

training set (data leakage), which would invalidate the distillation claims. The lack of a dataset hash or a public repository link for the 256K pairs makes the training data opaque.

****2. Evaluation Prompt Transparency:**** In the unnumbered paragraph of Section 5, the authors claim to evaluate on "200 evaluation prompts" using VBench. They assert that baseline scores were computed using the "identical VBench evaluation protocol (same prompts)." However, the specific list of these 200 prompts is not included in the text, nor is a link provided to the prompt set. Without access to the exact prompts, it is impossible to verify if the "fair comparison" holds or if the results are sensitive to prompt selection bias. This omission renders the quantitative results (Tables 1, 2, 3) unverifiable.

****3. Artifact Availability and Link Rot:**** The bibliography contains a critical failure in artifact availability. Reference 'c-002' (the model weights) is marked as "pending," and reference 'c-001' (PyTorch) is unreachable. The abstract claims code is released under an MIT license and weights under Apache 2.0, yet the primary artifact (the model) is not accessible. This constitutes a fatal data quality issue; without the weights, the "AnyFlow" model cannot be audited, and the reported VBench scores cannot be reproduced.

****4. Missing Data Versioning:**** The paper references "Wan2.1" and "Wan2.1-T2V-14B" but does not specify the exact commit hash or version tag of the base model used for distillation. Given the rapid iteration in diffusion models, using an unspecified version of the teacher model introduces significant ambiguity in the experimental setup.

To proceed, the authors must release the synthetic dataset (or a representative sample with a manifest), the exact evaluation prompt list, and ensure the model weights are hosted at a stable, accessible URL.

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paper_reviewer_figure_critic

Verdict: minor_revision

The manuscript presents a comprehensive set of figures, but several critical issues regarding accessibility, legibility, and format robustness require attention before publication.

First, ****accessibility is currently non-functional****. In 'preamble.tex', the command `\newcommand{\alttext}[1]{}{}` is defined as an empty macro. Consequently, the `\alttext{...}` placeholders inserted in the figure environments (e.g., 'figures/ablation_anyflow_simulation.tex') provide no actual description for screen readers. Every figure must be updated with a meaningful, descriptive string inside the `\alttext` command to comply with accessibility guidelines.

Second, ****legibility at print scale is compromised in Figure 5**** ('figures/abla-

tion_time_sampler.tex'). The table within this figure utilizes `\resizebox{\linewidth}{\height}`, which indiscriminately scales the content. This often results in axis labels or table text becoming too small to read when printed or viewed on standard screens. The authors should replace this with a fixed font size (e.g., `\small` or `\footnotesize`) and adjust column widths manually to ensure all text remains crisp and readable.

Third, **reliance on interactive video elements is risky**. Figures 1, 3, 4, 6, 7, and 8 use the `'animate'` package (`\animategraphics`) and explicitly instruct readers to "click and play" in Adobe Acrobat. This functionality is lost in web-based PDF viewers, mobile devices, and printed copies. To ensure the figures "earn their place" regardless of the viewing medium, the authors should include static keyframe comparisons (e.g., side-by-side frames at $t=0$, $t=10$, $t=20$) or high-quality GIFs embedded directly in the PDF. The current captions describe the video content well, but the visual evidence is inaccessible without the specific Adobe plugin.

Finally, **layout consistency in Figure 2** (`figures/ablation_interpolated_embedding.tex`) needs verification. The subfigures are set to `\textwidth`, which may cause the images to touch the margins or overlap with the sub-captions depending on the page layout engine. A manual check of the compiled PDF is required to ensure sufficient whitespace and that axis labels in `interpolated_embedding.pdf` are not clipped.

- [writing] The manuscript presents a comprehensive set of figures, but several critical issues regarding accessibility, legibility, and format robustness require attention before publication. First, accessibility is currently non-functional. In `preamble.tex`, the command `\newcommand{\alttext}[1]{}` is defined as an empty macro. Consequently, the `\alttext{...}` placeholders inserted in the figure environments (e.g., `figures/ablation_anyflow_simulation.tex`) provide no actual description for screen readers. Ever

paper_reviewer_jargon_police

Verdict: minor_revision

The manuscript exhibits a high density of specialized acronyms and jargon that significantly hinders accessibility for non-specialist readers. While the technical depth is appropriate for the field, the failure to define standard acronyms at their first occurrence violates basic readability principles.

Specifically, the term **NFEs** (Number of Function Evaluations) is defined in the Abstract but is used extensively in the Introduction and Method sections without reiteration or plain-language alternatives like "sampling steps." Similarly, **PF-ODE** (probability-flow ODE) appears in Section 3 without the full expansion, assuming the reader already knows the specific ODE formulation used in diffusion models.

In Section 2 and 3, **JVPs** (Jacobian-vector products) and **FSDP** (Fully Sharded Data Parallel) are used as if they are common knowledge, which excludes readers from adjacent fields or those new to large-scale training infrastructure. **DMD** (Distribution Matching Distillation) is introduced in Section 3.2 and 4.2 without the full phrase preceding

the acronym, forcing the reader to guess or search for the definition.

Furthermore, **CFG** (Classifier-Free Guidance), **KV-cache**, and the shorthand **T2V/I2V** are deployed without expansion. The text frequently relies on these abbreviations to save space, but this comes at the cost of clarity. For instance, "KV-cache size" in Section 4.3 is opaque to anyone not intimately familiar with transformer inference optimization.

To meet the standards of inclusive scientific communication, the authors must spell out every acronym at its first appearance in each major section or provide a glossary. Replacing jargon-heavy phrases with plain English equivalents (e.g., "sampling steps" instead of "NFEs" in the Introduction) would greatly improve the paper's reach without sacrificing precision.

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paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a coherent logical structure regarding the transition from consistency models to flow maps for any-step video generation. The premise that consistency models suffer from trajectory drift due to re-noising (Section 1, lines 25-30) logically supports the proposed solution of flow map transitions. The ablation studies in Table 4 (lines 145-165) generally support the claim that the proposed backward simulation improves multi-step performance compared to consistency baselines.

However, there are specific logical inconsistencies in the text that require clarification:

1. **Ethics Statement Logic**: The Ethics Statement (lines 48-62) presents a disjointed argument. It lists risks (deepfakes) and then immediately lists mitigation strategies. However, the sentence "We further note that AnyFlow builds upon..." is inserted between the risk acknowledgment and the mitigation list, breaking the logical chain. Furthermore, the mitigation strategies (watermarking) are proposed as future work or general suggestions but are not logically linked to the specific implementation details of AnyFlow in the current work, creating a gap between the problem statement and the proposed solution's actual scope.

2. **Experimental Protocol Consistency**: In Section 5 (lines 108-115), the authors claim to re-evaluate key counterparts using an "identical VBench evaluation protocol" to ensure fairness. Yet, the text immediately follows with "Results for all other methods are taken directly from their original papers." This creates a logical contradiction: if results are taken from original papers, the protocol (prompts, seeds, aggregation) likely differs. If

the protocol was identical, the authors must have re-run those specific baselines, which should be explicitly stated for **all** baselines, not just “key counterparts,” to support the fairness claim. The current phrasing leaves the validity of the comparison ambiguous.

3. **Causal Attribution**: The abstract and introduction attribute the failure of consistency models at high NFEs primarily to “trajectory drift” caused by re-noising. While the paper demonstrates that AnyFlow performs better, it does not strictly isolate “trajectory drift” as the **sole** causal factor versus other potential confounders (e.g., the specific nature of the flow map loss vs. consistency loss). The logical leap from “AnyFlow works better” to “Therefore, trajectory drift was the exclusive cause of consistency failure” is slightly overstated without a direct ablation of the drift mechanism itself.

These issues are primarily textual and logical clarifications rather than fundamental flaws in the methodology, warranting a minor revision.

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paper_reviewer_overreach

Verdict: minor_revision

The paper makes several strong claims regarding the novelty and statistical significance of the AnyFlow framework that appear to overreach the provided evidence.

First, the Abstract and Introduction repeatedly assert that AnyFlow is the “first any-step video diffusion distillation framework based on flow maps.” However, Section 2 (Related Work) explicitly cites a concurrent work, TMD (Nie et al., 2026), which “also studies a flow map formulation for bidirectional video diffusion distillation.” While the authors distinguish their work by focusing on “simulation efficiency” via shortcut decomposition

versus TMD’s architectural changes, the absolute claim of being the “first” framework is an over-claim. The novelty should be framed more precisely as the first to achieve *efficient* any-step distillation via flow map backward simulation, rather than the first to apply flow maps to video distillation generally.

Second, the manuscript makes a definitive statistical claim in the text preceding the Ethics Statement: “All reported improvements are evaluated with paired t-tests and Bonferroni correction to control the family-wise error rate.” This is a significant over-claim of rigor because the paper does not report the resulting p-values, the specific confidence intervals (beyond a generic mention), or the effect sizes. Given that many reported improvements are marginal (e.g., 84.05 vs 83.25, or 83.96 vs 83.49), asserting statistical significance without presenting the actual test statistics is misleading. The authors must either provide the statistical tables or soften the language to “we observed consistent improvements” rather than claiming formal significance testing was successfully applied and validated.

Third, there is a contradiction in the evaluation protocol that undermines the “fair comparison” claim. The text states, “Results for all other methods are taken directly from their original papers,” yet it also claims to “re-evaluate key counterparts... using the official VBench augmented prompts” for a “fair comparison.” Specifically, the comparison against Krea-Realtime-14B (83.25) appears to be a direct citation from another paper, while AnyFlow is evaluated on 600 videos (200 prompts x 3 seeds). If the prompt distributions or random seeds differ between the cited Krea score and the AnyFlow evaluation, the claim of a “fair comparison” is an overreach. The authors must clarify whether the Krea baseline was re-run or if the comparison is cross-study, and adjust the “fair comparison” language accordingly.

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paper_reviewer_safety_ethics

Verdict: minor_revision

The paper addresses significant safety and ethical concerns inherent to video diffusion models, specifically the risks of deepfakes and disinformation. The authors correctly identify these risks in the Ethics Statement (lines 48-62) and propose standard mitigation strategies such as watermarking, usage policies, and detection tools. However, the current presentation of these strategies is structurally flawed.

Specifically, in the Ethics Statement, the sentence “To mitigate these risks, we propose the following strategies:” is immediately followed by “We further note that AnyFlow builds upon...” without actually listing the strategies. The bulleted list containing Watermarking,

Usage Policies, and Detection Tools appears syntactically orphaned, making it unclear if these are the authors' proposals or a general observation. This grammatical disconnect undermines the clarity of their safety commitments and must be corrected to ensure the mitigation plan is explicitly attributed to the authors.

Furthermore, the claim regarding data provenance requires substantiation. The authors state that "synthetic training videos were generated from publicly available, royalty-free video clips with appropriate consent" (lines 53-54). In the context of generative AI, vague references to "publicly available" data are insufficient for ethical review. The authors must explicitly name the dataset(s) used, provide the specific license terms (e.g., CC-BY, Public Domain), and describe the mechanism by which "appropriate consent" was verified or assumed. Without this transparency, the claim of compliance with copyright and privacy regulations cannot be verified.

Finally, while the authors acknowledge that their current evaluation benchmark (VBench) lacks safety and fairness metrics (lines 43-45), the paper lacks a concrete roadmap for addressing this limitation. Merely stating that "Future work should incorporate dedicated benchmarks" is insufficient for a paper releasing a powerful generative model. The authors should clarify if they have performed any internal safety audits, bias checks, or red-teaming of the model outputs, and if so, report those findings or commit to releasing a safety evaluation protocol alongside the code.

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paper_reviewer_scientific_evidence

Verdict: minor_revision

The paper presents a novel methodology for any-step video diffusion, but the scientific evidence supporting the central claims of statistical superiority and robustness is currently incomplete.

The most critical gap is the lack of reported variance in the quantitative results. The text explicitly claims: "Scores are averaged over three random seeds; we report mean $\pm$ standard deviation and 95% confidence intervals in Table 4" and "All reported improvements are evaluated with paired t-tests." However, the provided LaTeX source contains no "Table 4," and the existing tables (e.g., 'tables/t2v_comparison.tex', 'tables/ablation_anyflow.tex') only report single-point mean scores. Without the standard deviations or the raw data, the claims of statistical significance (p-values) are unverifiable. The differences reported in the ablation studies (e.g., 83.54 vs 83.49 in Table 3) are minute; without variance estimates, these could easily be noise rather than genuine improvements.

Furthermore, the sample size for the primary evaluation (200 prompts) is on the lower end for video generation benchmarks, which are known to have high stochastic variance. While the authors mention using three seeds, the effective sample size for the prompt distribution is only 200. The paper should either increase the prompt count to ensure robust statistical power or provide a rigorous justification for why 200 prompts are sufficient to detect the reported effect sizes. The absence of confidence intervals in the main result tables prevents a proper assessment of the reliability of the "AnyFlow" improvements over baselines like rCM and Self-Forcing.

Finally, the training cost analysis in Table 5 ('tables/training_cost.tex') reports single measurements (e.g., "53.1 s/iter") without indicating the variance across iterations or runs. While less critical than the generation metrics, this lack of statistical reporting undermines the precision of the efficiency claims. To meet the standards of scientific evidence, the authors must include standard deviations for all reported metrics and provide the missing statistical tables or raw data to substantiate the t-test claims.

- [science] The claim of statistical significance (paired t-tests, Bonferroni correction) in the main text is unsupported by data. The paper reports mean scores but fails to provide the standard deviations or confidence intervals in the main tables (e.g., Table 1, Table 2) or the text, despite explicitly stating they are reported in 'Table 4' which does not exist in the provided source. Without the variance data, the t-tests cannot be verified.
- [science] The evaluation sample size for the primary downstream fine-tuning claim is insufficient to support broad generalization. The text states '200 evaluation prompts (one video per prompt) with three random seeds' (600 total). For high-variance generative models, 200 prompts is a small sample size for robust statistical inference. The paper must justify this N or provide a power analysis, and explicitly report the standard deviation across the 200 prompts, not just the seeds.
- [science] The ablation study in Table 3 (ablation_anyflow) presents results with two decimal places (e.g., 83.54 vs 83.49) but does not report the standard error or variance for these specific ablation runs. Given the small differences (e.g., 0.05 points), it is impossible to determine if these improvements are statistically significant or within the noise floor of the evaluation metric without the variance data.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The statistical analysis presented in Section 5 contains critical omissions that prevent the verification of the paper's central claims regarding performance improvements.

First, while the text explicitly states that "All reported improvements are evaluated with paired t-tests and Bonferroni correction" (Section 5, paragraph starting "Statistical significance"), the manuscript fails to provide the necessary statistical evidence. No p-values, t-statistics, or degrees of freedom are reported in the text or tables. Without these values, the assertion that the improvements are statistically significant is unsubstantiated.

The authors must report the specific p-values for the comparisons in Tables 1, 2, and 3 to validate the use of the t-test.

Second, there is a discrepancy between the text and the tables regarding uncertainty quantification. The text mentions reporting “mean $\pm$ standard deviation and 95% confidence intervals in Table 4” (likely referring to the main results tables, though the numbering is ambiguous). However, Tables ‘t2v_comparison’, ‘i2v_comparison’, and ‘ablation_anyflow’ present only single point estimates (e.g., 84.05, 87.87). The absence of standard deviations or confidence intervals in the tables makes it impossible to visually assess the overlap between distributions or the stability of the reported gains across the three random seeds.

Third, the experimental design for the main comparisons introduces a potential confound. The authors state that baseline scores for methods like rCM and Krea-Realtime were “taken directly from their original papers” (Section 5, ‘Evaluation Setting’), while AnyFlow was re-evaluated. Even if the prompt sets are identical, differences in random seeds, hardware environments, or implementation details between the original papers and the current re-evaluation violate the assumption of a controlled paired comparison required for a valid t-test. To ensure the statistical validity of the “outperforming” claims, all baselines must be re-evaluated under the exact same protocol (same seeds, same hardware, same prompt set) as AnyFlow.

Finally, the sample size of 200 prompts (600 total videos) is reasonable, but the power of the test depends on the variance of the VBench scores. Given that VBench scores are often bounded and potentially non-normally distributed, the authors should verify the normality assumption of the t-test or consider non-parametric alternatives (e.g., Wilcoxon signed-rank test) if the data distribution is skewed, and report the results accordingly.

- [science] The manuscript claims statistical significance via paired t-tests with Bonferroni correction (Section 5, ‘Statistical significance’ paragraph) but fails to report the resulting p-values, t-statistics, or degrees of freedom. Without these values, the claim of significance is unverifiable.
- [science] The evaluation protocol states 200 prompts with 3 seeds (600 videos) for VBench. However, the reported scores in Tables 1-3 are single point estimates (e.g., 84.05) without the standard deviation or 95% confidence intervals explicitly listed in the tables, despite the text claiming they are reported. The tables must include these variance metrics to assess the robustness of the improvements.
- [science] The comparison of AnyFlow against baselines (e.g., rCM, Krea-Realtime) mixes re-evaluated results with scores ‘taken directly from original papers’ (Section 5, ‘Evaluation Setting’). This introduces potential confounding variables (different hardware, random seeds, or prompt sets) that invalidate the paired t-test assumption of identical conditions. A unified re-evaluation of all baselines is required.

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript demonstrates a high level of technical sophistication, and the writing is generally clear and professional. However, there are specific structural and grammatical issues that disrupt the reading flow and require attention before final submission.

The most significant issue is found in the **Ethics Statement** section (arxiv_anyflow.tex). The text introduces a list of mitigation strategies ("To mitigate these risks, we propose the following strategies:") but is immediately interrupted by a paragraph discussing data provenance ("We further note that AnyFlow builds upon...") before the actual itemized list begins. This creates a jarring logical break. The sentence introducing the strategies should be immediately followed by the `\begin{itemize}` block, and the data provenance sentence should be moved to a separate paragraph or integrated into the introduction/methodology where data sources are typically discussed.

In **Section 3 (Preliminary)**, specifically in 'sections/3_preliminary.tex', the subsection header `\myPara{Differential Derivation Equation.}` includes a trailing period inside the command argument. In LaTeX, this often results in awkward spacing or punctuation errors in the final PDF. The period should be removed from the header text itself, as the formatting command handles the necessary punctuation or spacing.

Additionally, while the term **NFEs** is defined in the abstract, it appears frequently in the **Experiments** section (e.g., " 50×2 NFEs"). For a broader audience, ensuring the definition is reiterated or that the notation is explicitly explained in the main text (perhaps in the Implementation Details) would improve accessibility. The current usage is mathematically precise but slightly dense for readers unfamiliar with the specific distillation literature's shorthand.

Finally, the transition between the **Introduction** and the **Method** section is smooth, but the **Related Work** section could benefit from slightly tighter paragraph cohesion. Some transitions between the discussion of Consistency Models and Flow Map Models feel abrupt. A brief bridging sentence explaining *why* the shift from consistency to flow maps is necessary for the specific problem of "any-step" generation would enhance the narrative flow.

Overall, the paper is well-written, but these specific structural and punctuation errors should be corrected to ensure a polished final product.

- [writing] In the Ethics Statement (arxiv_anyflow.tex), the sentence 'We acknowledge that video diffusion models... To mitigate these risks, we propose the following strategies:' is immediately followed by a paragraph about data provenance before the itemized list begins. This breaks the logical flow. The list should immediately follow the proposal sentence, and the data provenance note should be moved to a separate paragraph or the Related Work section.
- [writing] In sections/3_preliminary.tex, the subsection header 'Differential Derivation Equation.' contains a period inside the header text, which is non-standard for LaTeX sectioning commands and looks like a typo. It should be 'Differential Derivation Equation'

without the trailing period.

- [writing] In sections/5_experiments.tex, the phrase '50\$ imes\$2 NFes' is used multiple times. While mathematically clear, the phrasing '50 steps with 2 frames' or '50 iterations of 2-frame generation' might be clearer for general readability, though the current notation is acceptable if defined earlier. Ensure 'NFes' is explicitly defined as 'Number of Function Evaluations' in the main text before this table, not just in the abstract.